

Alpha-Szegő Rank-One Dynamics: A Turning-Point-Safe Cascade Theorem

HCS-C386 theorem and reproducibility package

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Abstract

We give a turning-point-safe proof of the full rank-one threshold for the alpha-Szegő equation. Exact Fourier convolution preserves mass, momentum and energy, while an energy-defect estimate controls every off-threshold orbit. At positive perturbation, zero defect forces a strictly positive radial discriminant. Direct differentiation yields a polynomial second-order equation, so uniqueness remains valid at the point where radial velocity vanishes. Its solution is an explicit two-sided hyperbolic-secant-squared profile with an initial-data-dependent time shift. Non-relative-compactness is thus equivalent to the classical energy equality on the entire rank-one manifold. Zero perturbation, negative perturbation and spatial constants are handled separately; constants satisfy the equality but do not cascade. The proof does not infer trajectory uniqueness from a squared first-order identity, and it does not extend the threshold to higher ranks or arbitrary Hardy-space initial data.

中文摘要

本文闭合塞戈方程整个秩一流形上的正扰动级联阈值，并保留径向转折点的唯一性。

质量、动量与能量先给出平方速度恒等式，再从原方程直接推导二阶径向方程。

该二阶方程在速度为零处仍有效，排除了转折点停驻的伪解。

完整径向轨道为双向双曲正割平方型，时间平移由初始位置与速度唯一确定。

由此非相对紧性等价于经典能量阈值。零扰动、负扰动和秩零常数分别处理。

结论不推广到任意哈代空间初值或高秩流形，也不把经典阈值声称为新发现。

Keywords: alpha-Szegő; rank-one dynamics; energy threshold; turning point; Sobolev cascade; isospectrality

关键词：塞戈方程；秩一动力学；能量阈值；转折点；索伯列夫级联；等谱性

1 One invariant nonlinear source

A finite-rank invariant manifold need not be a finite Fourier approximation. Here it contains infinitely many modes and an exact nonlinear evolution. The cubic Szegő equation and its Hankel structure originate in Gérard–Grellier [1]; Xu [2] established the rank-one threshold for its linear perturbation. Our contribution within this repository is a complete convention-preserving proof and reproducibility package. In particular, we retain the rank hypothesis, the turning point, and the distinction between an auxiliary determinant and physical evolution. We do not claim literature priority for the classical threshold.

Use normalized circle measure and let Π retain nonnegative Fourier modes. For every real α fix

$$iu_t = \Pi(|u|^2 u) + \alpha(u \mid 1), \quad u(z) = b + \frac{cz}{1-pz}, \quad c \neq 0, \quad |p| < 1. \quad (1)$$

The phase space in (1) is $\mathcal{L}(1)$; constants will be treated separately. The physical time is t , without rescaling. Set

$$d = 1 - |p|^2, \quad \rho = |c|, \quad B = |b|^2, \quad Q = B + \frac{\rho^2}{d}, \quad M = \frac{\rho^2}{d^2} > 0. \quad (2)$$

Energy and its defect are

$$E_\alpha = \frac{1}{4} \|u\|_4^4 + \frac{\alpha}{2} |b|^2, \quad \delta = E_\alpha - \frac{Q^2}{4} - \frac{\alpha Q}{2}. \quad (3)$$

We always take the positive square root of M . On a trajectory, $\rho = \sqrt{M}d$ and $B = Q - Md$.

Theorem 1 (Exact all-mode reduction and global existence). *Every initial datum in $\mathcal{L}(1)$ has a unique solution for all real time. Its coefficients satisfy*

$$\begin{aligned} i\dot{b} &= (B + 2\rho^2/d + \alpha)b + \rho^2 c\bar{p}/d^2, \\ i\dot{c} &= (2B + \rho^2/d^2)c + 2b\rho^2 p/d, \\ i\dot{p} &= c\bar{b} + \rho^2 p/d. \end{aligned} \quad (4)$$

The quantities Q, M, E_α are conserved. These statements hold at $b = 0$ and $p = 0$ and involve no Fourier cutoff.

Proof. The geometric Fourier coefficients are $u_0 = b$ and $u_n = cp^{n-1}$ for $n \geq 1$. Their absolute convergence on strict subdisks justifies convolution in $\Pi(|u|^2 u)$. Compare the coefficients of $1, z/(1-pz), z^2/(1-pz)^2$; the derivative of the last coordinate is $c\dot{p}$, not merely $\dot{p}$. The result is (4). A direct quartic sum gives

$$4E_\alpha = B^2 + \frac{4B\rho^2}{d} + \frac{\rho^4(1+|p|^2)}{d^3} + \frac{4\rho^2\Re(bp\bar{c})}{d^2} + 2\alpha B. \quad (5)$$

Differentiating (2) and (5) along (4) cancels every term, giving the three conservation laws.

To justify all-time continuation, put $X = bp\bar{c}$. Direct differentiation gives

$$\dot{d} = 2\Im X, \quad \dot{\rho} = 2\sqrt{M}\Im X, \quad |\dot{\rho}/\rho| \leq 2\sqrt{QM}. \quad (6)$$

Hence $\rho(t) \geq \rho(0)e^{-2\sqrt{QM}|t|}$ on every finite time interval. The conserved quantities bound $|b| \leq \sqrt{Q}, \rho \leq \sqrt{M}$, and $d = \rho/\sqrt{M}$ away from zero on that interval. The real six-dimensional vector field is locally Lipschitz on $d > 0$, so its continuation theorem excludes finite-time exit. The resulting geometric series solves (1) in every H^s . This argument proves the named invariant-manifold theorem, not global well-posedness for arbitrary Hardy data. $\square$

2 Compactness and the exact exceptional faces

The defect is useful because it vanishes precisely where a compactness estimate could fail.

Lemma 1 (Factored defect and a quantitative bound). *For every trajectory in $\mathcal{L}(1)$,*

$$\delta = \frac{Md}{2} \left(\left| b + \frac{c\bar{p}}{d} \right|^2 - \alpha \right). \quad (7)$$

If $\delta \neq 0$, then for every real t ,

$$d(t) \geq \frac{2|\delta|}{M((\sqrt{Q} + \sqrt{M})^2 + |\alpha|)} > 0. \quad (8)$$

The orbit is relatively compact in $\mathcal{L}(1)$ and bounded in every H^s .

Proof. Subtract the two terms in (3) from (5); the remaining quadratic expression is (7). Since $|b + c\bar{p}/d| \leq \sqrt{Q} + \sqrt{M}$, taking absolute values proves (8). Thus $|p|$ lies in a strict subdisk and $|c| = \sqrt{M}d$ stays positive, while b, c remain bounded. These inequalities give compact closure within the coefficient manifold. Every Sobolev norm is then bounded by a convergent geometric series. $\square$

For $\alpha < 0$, (7) has strictly positive defect; thus all rank-one orbits are relatively compact. On a compact set of initial data, the positive continuous defect has a positive minimum, making the bound uniform for that set. This does not assert uniformity across data approaching a positive-alpha zero-defect hypersurface.

For $\alpha = 0$, nonzero defect is already settled. Zero defect forces $b = -c\bar{p}/d$. Writing $A = c/d$ yields

$$Q = M = |A|^2, \quad u(z) = A \frac{z - \bar{p}}{1 - pz}, \quad u(t, z) = e^{-iQt} u(0, z). \quad (9)$$

The quotient in (9) has modulus one on the circle. Substitution in (4) gives $\dot{p} = 0, \dot{b} = -iQb$ and $\dot{c} = -iQc$, proving the formula even at $p = 0$. Therefore every zero-alpha rank-one orbit is also relatively compact.

Rank zero cannot inherit the threshold. If $c = 0$, the parameter p is redundant and the complete solution is

$$u(t, z) = b(0)e^{-i(|b(0)|^2 + \alpha)t}. \quad (10)$$

Every constant satisfies $\delta = 0$ but has constant Sobolev norm. The zero function is stationary. A nonzero constant is stationary exactly when $|b|^2 + \alpha = 0$; otherwise its least temporal period is $2\pi/||b|^2 + \alpha|$. These facts explain why the rank-one hypothesis cannot be dropped from a positive-alpha growth criterion.

3 Turning-point-safe cascade

Assume $\alpha > 0$ and $\delta = 0$. Define the discriminant

$$\kappa^2 = 4QM - (\alpha - Q - M)^2. \quad (11)$$

The next result strengthens the usable form of the classical threshold: it provides the complete radial trajectory with a valid turning-point proof.

Theorem 2 (Complete positive-alpha dichotomy). *For initial data in $\mathcal{L}(1)$ and $\alpha > 0$, the following are equivalent: the trajectory is not relatively compact in $\mathcal{L}(1)$; and $\delta = 0$. On that level $\kappa > 0$, and, with*

$$d_* = \frac{\kappa^2}{4\alpha M}, \quad t_* = \frac{2}{\kappa} \operatorname{artanh} \left(\frac{\dot{d}(0)}{\kappa d(0)} \right), \quad (12)$$

the exact radial orbit for every real t is

$$d(t) = d_* \operatorname{sech}^2 \left(\frac{\kappa(t - t_*)}{2} \right), \quad \rho(t) = \sqrt{M} d(t). \quad (13)$$

Here $0 < d_* \leq 1$, and the formula includes $p = 0$ at the turn.

Proof. Equation (7) gives $2\Re X = d(\alpha - Q - M) + 2Md^2$. Together with $|X|^2 = (Q - Md)Md^2(1 - d)$ and (6), it implies

$$\dot{d}^2 = d^2(\kappa^2 - 4\alpha Md). \quad (14)$$

Since $d > 0$, the right side forces $\kappa^2 \geq 4\alpha Md > 0$. Moreover,

$$\kappa^2 - 4\alpha M = -(Q - M - \alpha)^2 \leq 0, \quad (15)$$

so $d_* \leq 1$.

A squared velocity identity alone permits spurious sticking at a turn. To remove that ambiguity, differentiate the complex ODE itself. Put $T = \alpha - Q - M$. Then

$$\begin{aligned} \dot{X} &= i(FX + G), \quad F = -T - 4Md, \\ G &= -M^2 d^2(1 - d) - (Q - Md)Md^2 + 2(Q - Md)Md(1 - d). \end{aligned} \quad (16)$$

All coefficients F, G are real. Since $\dot{d} = 2\Im X$, substitute $\Re X = dT/2 + Md^2$ in $\ddot{d} = 2(F\Re X + G)$ to obtain the polynomial equation

$$\ddot{d} = \kappa^2 d - 6\alpha Md^2. \quad (17)$$

This derivation never divides by $\dot{d}$ or by p . At $d = d_*$ it gives $\ddot{d} = -\kappa^2 d_*/2 < 0$, excluding a stationary turning point.

By (14), $v_0 = \dot{d}(0)/(\kappa d(0))$ lies strictly between -1 and 1 , so (12) is finite. The function (13) solves (17). At time zero its logarithmic derivative is $\kappa \tanh(\kappa t_*/2) = \kappa v_0$. Also $\operatorname{sech}^2(\kappa t_*/2) = 1 - v_0^2 = d(0)/d_*$, so both initial data agree. Uniqueness for the polynomial second-order initial-value problem proves (13) for every real t , including the turn.

The radial variable tends to zero at both infinite-time ends, so this orbit does not have compact closure within $\mathcal{L}(1)$. Every nonzero-defect orbit is compact by (8). This proves both directions of the dichotomy. $\square$

The all-time clause matters. No finite-time singularity is created: $d(t)$ remains positive for every finite t . The expression $\kappa^2/(4\alpha M)$ can equal one, but that is the harmless event $p = 0$, not the forbidden pole boundary $|p| = 1$.

The source threshold appears in Xu [2, Theorem 3.1]. We do not rely on a globally positive lower bound for $|\dot{d}/d|$: that derivative is zero at $t = t_*$. Equations (16)–(17) and uniqueness provide the complete argument instead.

4 Evidence boundary and conclusion

Source face	Defect	Established behavior
Rank one, $\alpha < 0$	positive	relatively compact
Rank one, $\alpha = 0$	nonzero	relatively compact
Rank one, $\alpha = 0$	zero	inner-function phase orbit
Rank one, $\alpha > 0$	nonzero	relatively compact
Rank one, $\alpha > 0$	zero	exact two-sided cascade
Rank zero	always zero	constant-norm phase orbit

The regime table records a rank distinction, not a fitted classifier. All complex phases are retained in the coefficient ODE. The all-time and all-frequency conclusions are proved analytically; finite computations test independent reconstructions of their formulas.

The claim remains the rank-one source theorem. No arbitrary Hardy-data or higher-rank threshold, complete generic primitive census, target arithmetic local data, target Euler factors, root number, automorphy, target divisor, target functional equation, target-zero matching, or Hilbert–Pólya operator is claimed. Route B remains disabled. The firewall is NO_BAD_EULER_OR_ROOT_NUMBER.

Literature ownership and reproducibility. The classical equation and threshold belong to the cited sources. Our package makes the turning-point argument, rank-zero failure, physical clock and auxiliary-determinant limitation auditable together. No publication acceptance, worldwide novelty, external-human review or cross-provider validation is asserted. AI-assisted local drafting and independent collaborating-agent review are disclosed. The source proof, locked evaluation, independent scripts and three substantive PDF revisions accompany the article.

Conclusion. Round one: turning-point-safe cascade theorem. The positive-alpha energy equality gives the exact two-sided radial law; the direct acceleration equation excludes an artificial stationary turn.

References

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